

Combinatorics of Quiver Mutations

Scott Neville

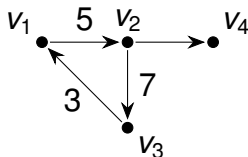
Department of Mathematics
University of Oregon

Sumner Myers Colloquium,
September 2026

partially based on joint work with
Sergey Fomin

Definition

A **quiver** is a finite directed graph. Multiple arrows are allowed, but not loops nor oriented 2-cycles. The arrow multiplicities are called **weights**.

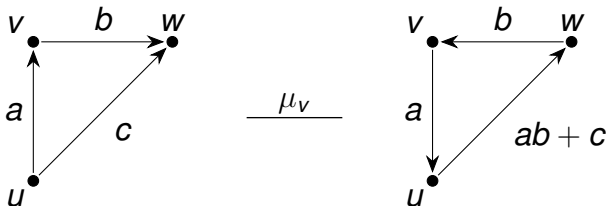


Quiver mutation

Definition

The quiver **mutation** $\mu_v(Q) = Q'$ is computed in three steps.

- If $u \xrightarrow{a} v \xrightarrow{b} w$ then add $a \cdot b$ arrows $u \longrightarrow w$.
- Reverse all arrows incident to v .
- Delete oriented 2-cycles, one by one.



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Proposition

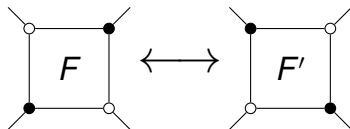
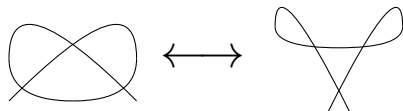
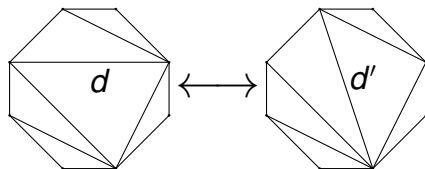
Mutations are involutions: $\mu_v(\mu_v(Q)) = Q$.

Definition

Two quivers are **mutation equivalent** if they are related by a sequence of mutations.

Motivation for quiver mutation

$$x^2 + y^2 + z^2 = xyz$$



Motivation 1: Markov triples

Definition [A. Markov 1879]

A **Markov triple** is a positive integer solution (x, y, z) to the equation

$$x^2 + y^2 + z^2 = xyz.$$

Motivation 1: Markov triples

Definition [A. Markov 1879]

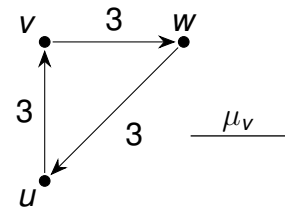
A **Markov triple** is a positive integer solution (x, y, z) to the equation

$$x^2 + y^2 + z^2 = xyz.$$

One such triple is $(3, 3, 3)$. There are many more.

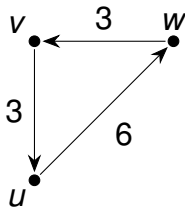
Markov triples from the Seiberg quiver

$$x^2 + y^2 + z^2 = xyz.$$



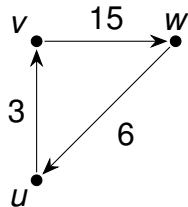
Seiberg Quiver

μ_v



$(3, 3, 6)$

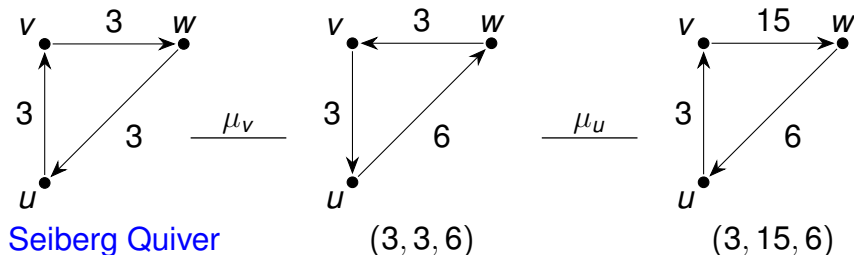
μ_u



$(3, 15, 6)$

Markov triples from the Seiberg quiver

$$x^2 + y^2 + z^2 = xyz.$$



Proposition [folklore]

A triple of positive integers (x, y, z) is a Markov triple if and only if it is a triple of weights appearing in a quiver mutation equivalent to the Seiberg quiver.

The Markov unicity conjecture

Conjecture [G. Frobenius 1913]

Suppose that (x, y, z) is a Markov triple with $x \leq y \leq z$.
Then z uniquely determines x and y .

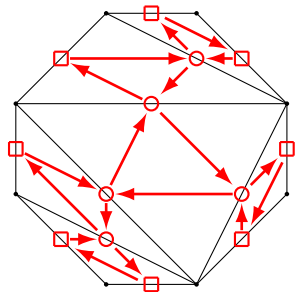
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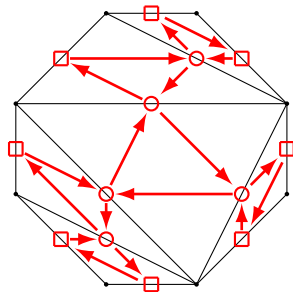
Suppose that (x, y, z) is a Markov triple with $x \leq y \leq z$.
Then z uniquely determines x and y .

Substantial progress has been made on this problem using
[cluster algebras](#) [K. Lee-L. Li-M. Rabideau-R. Schiffler 2023].

Movivation 2: Triangulations and their quivers



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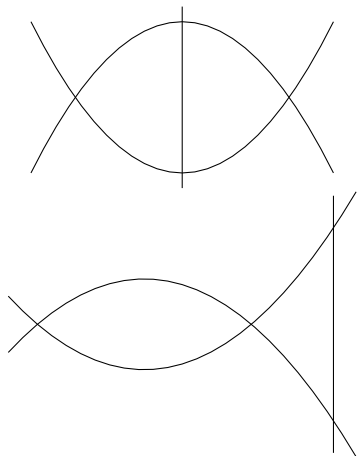


Theorem [W. Gu 2012]

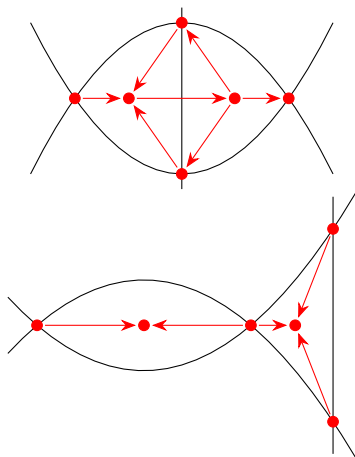
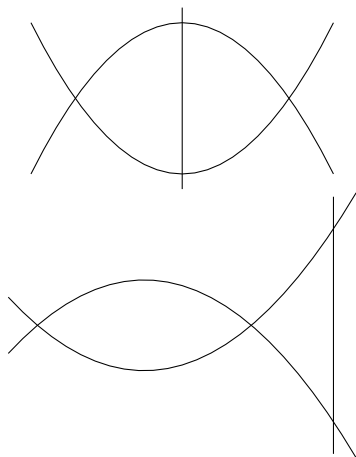
Fix two triangulations of Riemann surfaces with boundary and marked points. The following are equivalent:*

- the two triangulations come from the same marked surface;
- the associated quivers are mutation equivalent.

Motivation 3: Nodal curves and their quivers



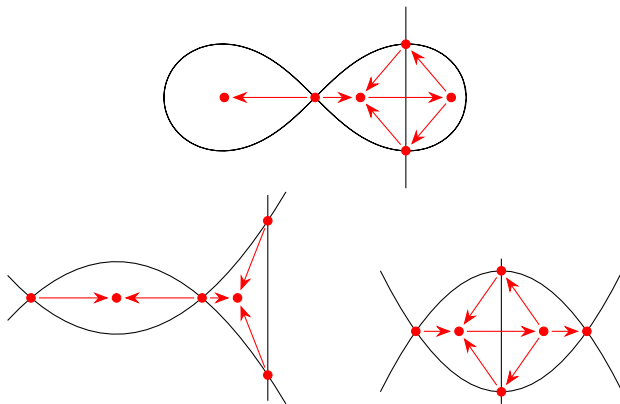
Motivation 3: Nodal curves and their quivers



[N. A'Campo, S. Gusein-Zade, S. Fomin]

Beyond triangle moves

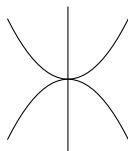
These three **quivers** are mutation equivalent.



What do these curves have in common?

Motivation 4: Singularities

Consider two real forms of the same complex singularity:



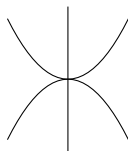
$$x(y^2 - x^4) = 0$$



$$x(y^2 + x^4) = 0$$

Motivation 4: Singularities

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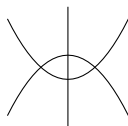


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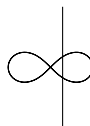


$$x(y^2 + x^4) = 0$$

After a perturbation (creating a [morsification](#)):

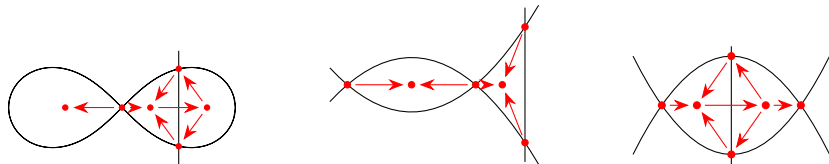


$$x(y^2 - (x^2 - \epsilon)^2) = 0$$



$$(x - \epsilon)(y^2 + x^4 - \epsilon(x^2 - y^2)) = 0$$

Morsifications and their quivers

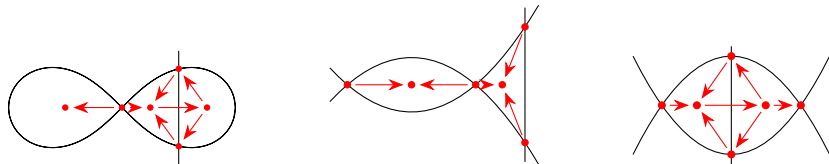


Conjecture [S. Fomin-P. Pylyavskyy-E. Shustin-D. Thurston 2017]

Given two morsifications of real isolated plane curve singularities (C_1, p_1) and (C_2, p_2) , the following are equivalent:

- (a) (C_1, p_1) & (C_2, p_2) have the same complex topological type;
- (b) the associated quivers are mutation equivalent.

Morsifications and their quivers



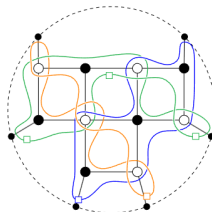
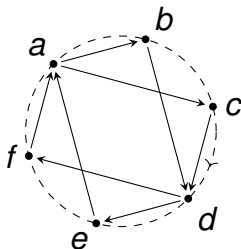
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The implication (b) $\implies$ (a) has been proven by [R. Casals 2026], under a technical assumption on morsifications.

Two UM theses



These results are closely related to two recent UM theses:
[SN 2025] and [A. Schwartz 2026].

Main problems

Big open question

Find an algorithm that detects mutation equivalence.

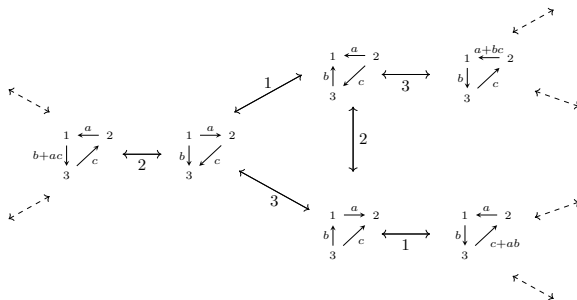
Main problems

Big open question

Find an algorithm that detects mutation equivalence.

Definition

The **mutation graph** of a quiver Q is the graph whose vertices are labeled by quivers in the mutation class of Q , and whose edges correspond to mutations.



Big open question

Find an algorithm that detects mutation equivalence.

- Problem 1: Understand the structure of mutation graphs.
- Problem 2: Find powerful invariants of quiver mutations.

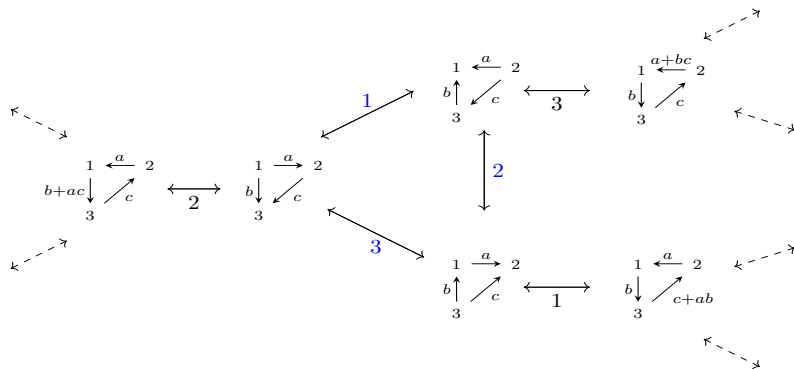
Long mutation cycles

S. Fomin–SN, Long mutation cycles, *Selecta Math.* **31** (2025).

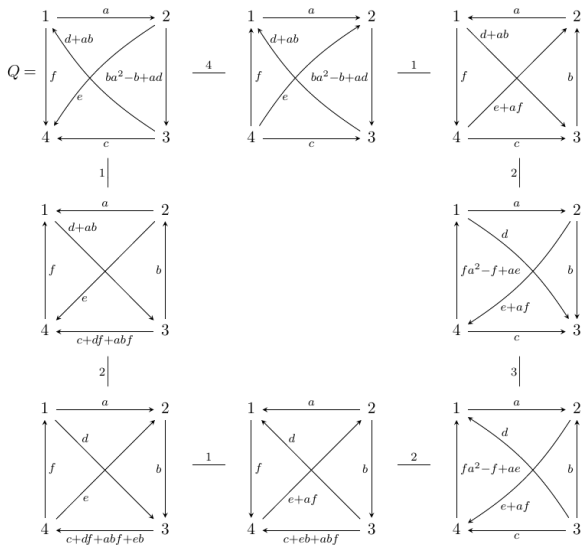
Mutation cycles

Definition

A **mutation cycle** is a cycle in the mutation graph.



Example of a new mutation cycle



$$a, b, c, d, e, f \geq 2$$

Long mutation cycles theorem

Theorem

For any $n \geq 4$ and $k \geq 1$, there exists a family $\mathcal{F}_{n,k}$ of n -vertex quivers such that:

- $\mathcal{F}_{n,k}$ is parameterized by $\binom{n}{2}$ positive integer parameters;
- each quiver in $\mathcal{F}_{n,k}$ lies on a mutation cycle of length $4k + n$;
- this mutation cycle cannot be paved by mutation cycles of length $\leq 4k$.

Mutation invariants

S. Fomin–SN, Cyclically ordered quivers, *Adv. Math.* **500** (2026).

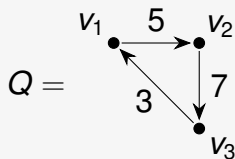
SN, Mutation-acyclic quivers are totally proper, arXiv:2409.17832.

Known mutation invariants

Definition

A quiver Q on a linearly ordered set of n vertices is encoded by an $n \times n$ **exchange matrix** B_Q .

Example



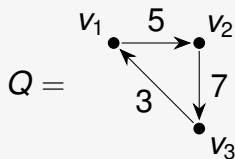
$$B_Q = \begin{pmatrix} 0 & 5 & -3 \\ -5 & 0 & 7 \\ 3 & -7 & 0 \end{pmatrix}$$

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Two known mutation invariants:

- The rank of the exchange matrix B_Q .
- The determinant of the exchange matrix B_Q .

Definition

Two $n \times n$ integer matrices M and M' are **congruent over $\mathbb{Z}$** if there exists a matrix $G \in \text{GL}(n, \mathbb{Z})$ such that $M' = GMG^T$.

The **integral congruence class** of an $n \times n$ integer matrix M consists of all matrices congruent to M over $\mathbb{Z}$.

Integral congruence

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Lemma

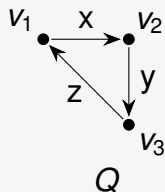
The integral congruence class of the exchange matrix is invariant under mutations.

The unipotent companion

Definition

The **unipotent companion** of B_Q is the unipotent upper-triangular matrix $U = U_Q$ such that $U^T - U = B_Q$.

Example



$$\begin{pmatrix} 0 & x & -z \\ -x & 0 & y \\ z & -y & 0 \end{pmatrix}$$

 B_Q

$$\begin{pmatrix} 1 & -x & z \\ 0 & 1 & -y \\ 0 & 0 & 1 \end{pmatrix}$$

 U_Q

Alexander polynomial

Definition

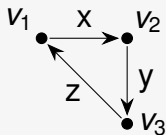
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Definition

The **Alexander polynomial** of B_Q is defined by

$$\Delta_Q(t) = \det(tU - U^T).$$

Example

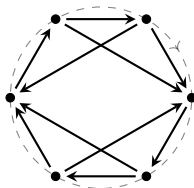


$$U = \begin{pmatrix} 1 & -x & z \\ 0 & 1 & -y \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Delta_Q(t) = (t-1)^3 - (x^2 + y^2 + z^2 - xyz)(t^2 - t).$$

Proposition

The integral congruence class of U is preserved by **cyclic shifts**.

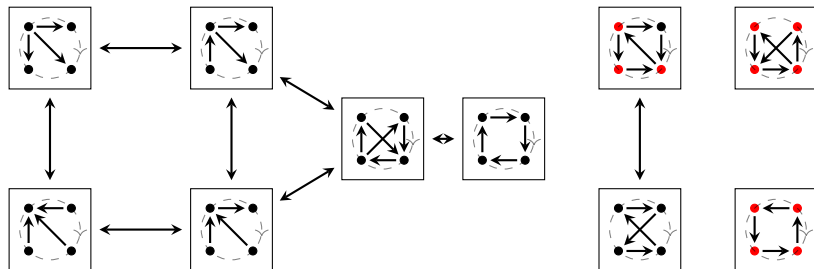


Definition

A **cyclically ordered quiver** is a quiver equipped with a cyclic ordering of the vertices.

Preservation of invariants

Mutations at "proper" vertices of a cyclically ordered quiver Q preserve the integral congruence class of U_Q .



proper mutation class of a cyclically ordered quiver Q



integral congruence class of U_Q



$\mathrm{GL}(n, \mathbb{Z})$ conjugacy class of $U_Q^{-T} U_Q$



Alexander polynomial $\Delta_Q(t)$

Totally proper cyclically ordered quivers

Observation

If a cyclically ordered quiver is **totally proper**, then any sequence of mutations preserves all of our mutation invariants.

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Theorem

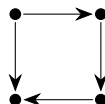
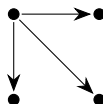
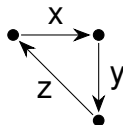
Any quiver possesses at most one totally proper cyclic ordering.

Examples of totally proper Cyclically ordered quivers

Theorem

The following types of quivers possess a totally proper cyclic ordering:

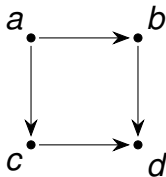
- any quiver on 3 vertices;
- any quiver of finite or affine type;
- 9 out of the 11 exceptional finite mutation types (cf. [H. Derksen–T. Owen 2008]).



Acyclic quivers are totally proper

Definition

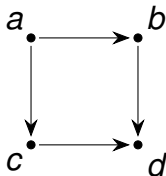
A quiver is **mutation-acyclic** if it is mutation equivalent to an **acyclic** quiver.



Acyclic quivers are totally proper

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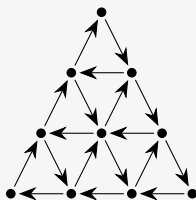
Theorem [SN 2024]

Every mutation-acyclic quiver is totally proper.

Example: Keller's quiver

Proposition

Keller's quiver



is not mutation-acyclic.

Proof

$$\Delta(t) = t^{10} - t^9 + t^7 - t^6 - t^4 + t^3 - t + 1$$

Any acyclic quiver with $\Delta(t) = t^{10} - t^9 + \dots$ is an orientation of a tree. There are 106 trees on 10 vertices. None of the associated Alexander polynomials match.

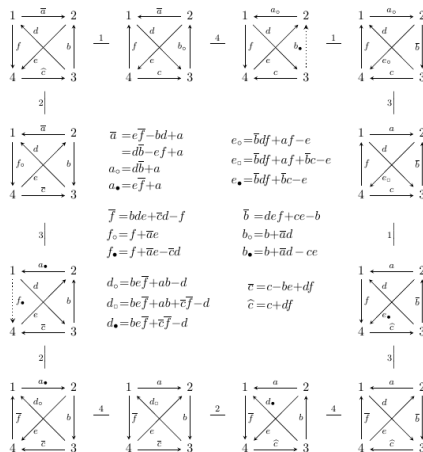
These ideas and results have been used to:

- disprove the mysterious points conjecture of [M. Castronovo–E. Gorsky–J. Simental–D. Speyer 2024], see [SN–J. Simental 2026];
- prove the generic case of the eventual sign coherence conjecture of [M. Gekhtman–T. Nakanishi 2019], see [A. Burcroff–SN 2026].
- build a database of small quivers with their invariants, for use in machine learning, see [T. Ervin–SN 2026+];
- extend the results on cyclically ordered quivers to skew-symmetrizable matrices, see [SN–H. Thomas 2026+].

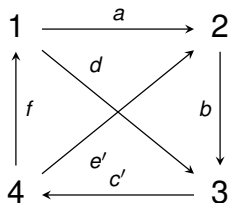
Thank you!

Even more cycles

We have constructed many other mutation cycles of 4-vertex quivers.



A different mutation cycle of length 12



Pick integers $a, b, c, d, e, f \geq 2$ (with $d > cf, e > af$). Set

$$e' = e + bc - af,$$

$$c' = be + b^2c - c.$$

The quiver above is on a mutation cycle with mutation sequence

123234341412